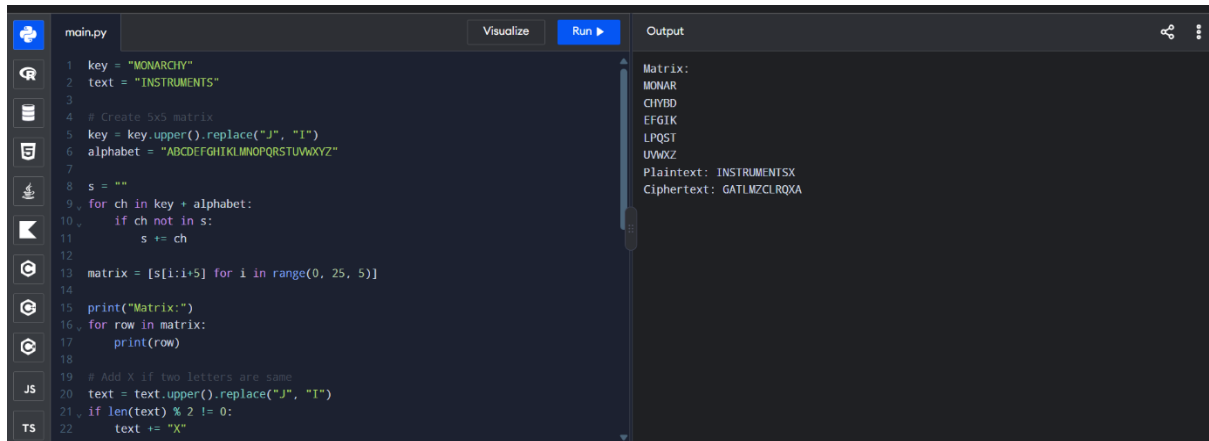


EXPERIMENT-3

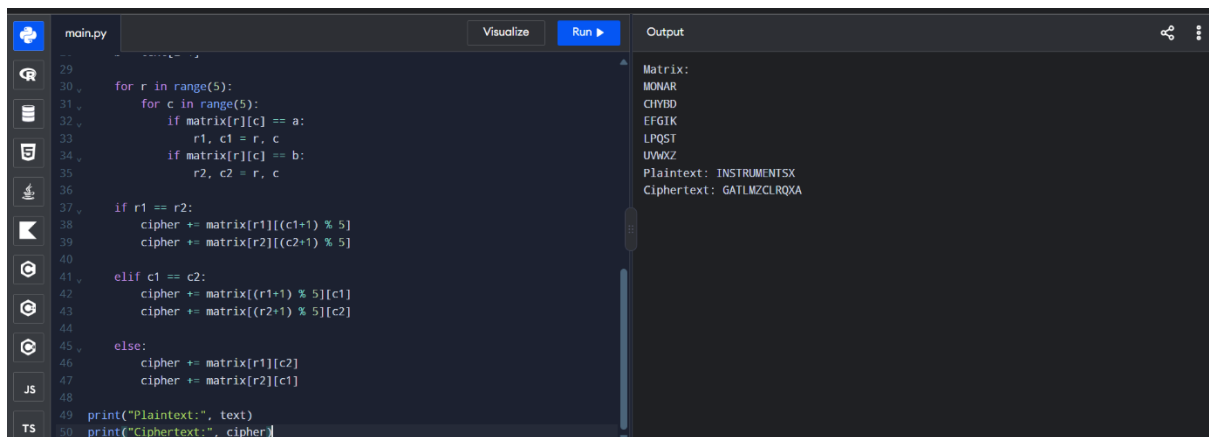
3. Write a C program for Playfair algorithm is based on the use of a 5 X 5 matrix of letters constructed using a keyword. Plaintext is encrypted two letters at a time using this matrix.



```
1 key = "MONARCHY"
2 text = "INSTRUMENTS"
3
4 # Create 5x5 matrix
5 key = key.upper().replace("J", "I")
6 alphabet = "ABCDEFGHIKLMNOPQRSTUVWXYZ"
7
8 s = ""
9 for ch in key + alphabet:
10     if ch not in s:
11         s += ch
12
13 matrix = [s[i:i+5] for i in range(0, 25, 5)]
14
15 print("Matrix:")
16 for row in matrix:
17     print(row)
18
19 # Add X if two letters are same
20 text = text.upper().replace("J", "I")
21 if len(text) % 2 != 0:
22     text += "X"
```

Output:

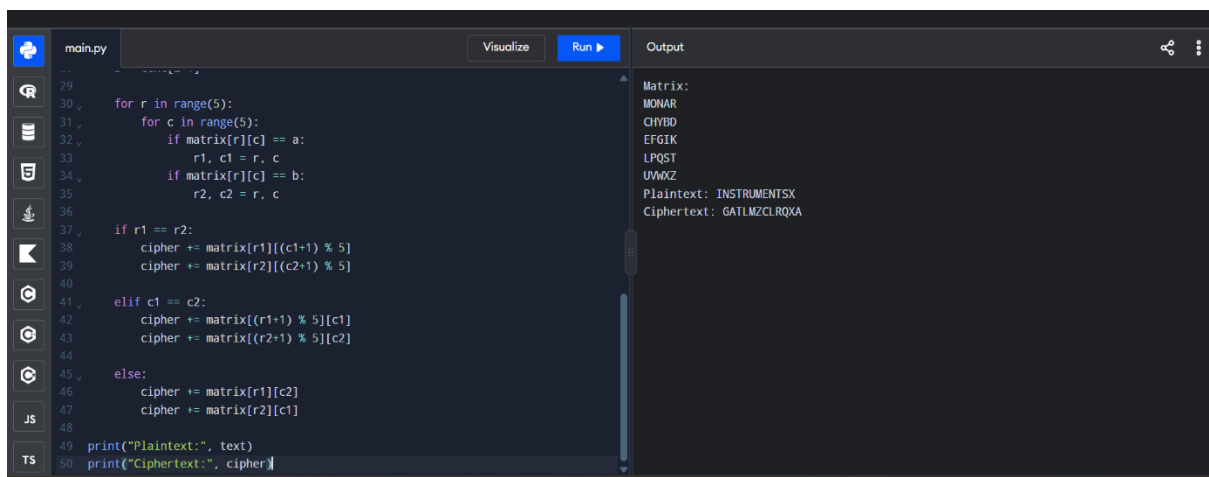
```
Matrix:
MONAR
CHYBD
EFGIK
LPQST
UWXXZ
Plaintext: INSTRUMENTSX
Ciphertext: GATLMZCLRQXA
```



```
29
30 for r in range(5):
31     for c in range(5):
32         if matrix[r][c] == a:
33             r1, c1 = r, c
34         if matrix[r][c] == b:
35             r2, c2 = r, c
36
37     if r1 == r2:
38         cipher += matrix[r1][(c1+1) % 5]
39         cipher += matrix[r2][(c2+1) % 5]
40
41     elif c1 == c2:
42         cipher += matrix[(r1+1) % 5][c1]
43         cipher += matrix[(r2+1) % 5][c2]
44
45     else:
46         cipher += matrix[r1][c2]
47         cipher += matrix[r2][c1]
48
49 print("Plaintext:", text)
50 print("Ciphertext:", cipher)
```

Output:

```
Matrix:
MONAR
CHYBD
EFGIK
LPQST
UWXXZ
Plaintext: INSTRUMENTSX
Ciphertext: GATLMZCLRQXA
```



```
29
30 for r in range(5):
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32         if matrix[r][c] == a:
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35             r2, c2 = r, c
36
37     if r1 == r2:
38         cipher += matrix[r1][(c1+1) % 5]
39         cipher += matrix[r2][(c2+1) % 5]
40
41     elif c1 == c2:
42         cipher += matrix[(r1+1) % 5][c1]
43         cipher += matrix[(r2+1) % 5][c2]
44
45     else:
46         cipher += matrix[r1][c2]
47         cipher += matrix[r2][c1]
48
49 print("Plaintext:", text)
50 print("Ciphertext:", cipher)
```

Output:

```
Matrix:
MONAR
CHYBD
EFGIK
LPQST
UWXXZ
Plaintext: INSTRUMENTSX
Ciphertext: GATLMZCLRQXA
```